EmpathAI — Mental Wellness Tracker

Java OOP Term II Project

A smart tracking system designed to help students and professionals detect early signs of burnout through daily behavioral monitoring.

Problem Statement & Vision

The Challenge

Students and professionals managing demanding workloads often overlook subtle signs of burnout until it's too late. Without simple, accessible daily tracking tools, stress patterns remain invisible until they escalate into serious issues.

Our Solution

EmpathAI provides an intuitive behavioral tracking system that captures sleep patterns, work sessions, and stress levels, transforming raw data into actionable insights about mental wellness.

Target Users

Students, developers, and working professionals managing stress and workload

Developer

Ahona Sarkar | IEM Gurukul | March 2025

System Architecture Overview

01

BehaviorSignal (Abstract Base)

Defines common contract for all behavioral data types

02

Concrete Signal Classes

SleepSignal, WorkSession, and StressEntry extend base class with specific implementations

03

DailyRecord Aggregator

Collects and organizes multiple signals for each day

04

User Profile

Stores complete history of daily records with controlled access

05

Service Layer

Analyzer computes stress scores while ReportGenerator produces insights

06

FileManager

Handles data persistence and file operations with error handling

Package Structure & Organization

model Package

Core entities including User, DailyRecord, and all BehaviorSignal implementations

service Package

Analyzer class for stress computation and ReportGenerator for insights generation

utils Package

FileManager handles persistence and provides helper utilities for data management

app Package

MainApp contains execution logic and user interface for system interaction

Object-Oriented Programming Concepts Applied

Abstraction

BehaviorSignal defines a common contract that all signal types must implement, ensuring consistent interface across different behavioral data.

Inheritance

SleepSignal, WorkSession, and StressEntry extend the abstract BehaviorSignal base class, inheriting common properties while adding specific behaviors.

Polymorphism

getStressContribution() method behaves differently for each signal type, allowing dynamic behavior based on the specific signal being processed.

Encapsulation

Private fields within User class are protected with controlled access through getter and setter methods, maintaining data integrity and security.

Composition

User contains DailyRecord objects, which in turn contain BehaviorSignal objects, creating a clear hierarchical relationship between system components.

Interface

Reportable interface implemented by ReportGenerator ensures consistent report generation contract across different reporting strategies.

Key Technical Highlights

Modular Signal-Based Scoring

Each signal independently contributes to stress score via getStressContribution(), ensuring extensibility and low coupling.

Method Overloading

Multiple generateReport() versions provide flexibility in report generation based on different parameters and requirements.

Exception Handling

Comprehensive error handling for file operations and invalid input validation prevents system crashes and maintains reliability.

Design Decision: Extensible Signal Architecture

Why It Matters: This approach allows adding new signal types (exercise, mood, nutrition) without modifying existing code. Low sleep generates high stress weight, high work hours add incremental stress, and direct stress input scales proportionally. The result is a flexible, scalable, and maintainable system that can evolve with user needs.

System Demonstration

Sample Weekly Report Output

```
2025-03-27 [HIGH]
2025-03-28 [LOW]
2025-03-29 [MODERATE]

Average score: 6.4 / 12
Recommendation: Monitor patterns and prioritize rest.
```

Key Features in Action

- Interactive daily data input for sleep, work, and stress levels
- Automated weekly report generation with visual stress indicators
- Color-coded risk level display for quick pattern recognition
- Personalized recommendations based on computed stress scores

Robust Error Handling

- Input validation prevents invalid stress values (e.g., stress >10)
- File operation recovery with warning messages instead of crashes
- Graceful data recovery from corrupted or missing files

```
User     : Ahona Sarkar
Age      : 20
User ID  : u001
Total days tracked : 4
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DAILY BREAKDOWN (last 7 days)
-----------------------------------
20025-03-25  ------  [HIGH]      (3 signals)
             └ [SLEEP | 20025-03-25] Slept 4.5 hours
             └ [WORK | 20025-03-25] Worked 10.0 hours
             └ [STRESS | 20025-03-25] Self-reported stress: 8/10
2025-03-26   -----------  [LOW]       (3 signals)
             └ [SLEEP | 2025-03-26] Slept 6.0 hours
             └ [WORK | 2025-03-26] Worked 7.5 hours
             └ [STRESS | 2025-03-26] Self-reported stress: 5/10
2025-03-27   ---------  [HIGH]      (3 signals)
             └ [SLEEP | 2025-03-27] Slept 3.5 hours
             └ [WORK | 2025-03-27] Worked 11.0 hours
             └ [STRESS | 2025-03-27] Self-reported stress: 9/10
2025-03-29   ------  [MODERATE]  (3 signals)
             └ [SLEEP | 2025-03-29] Slept 5.5 hours
             └ [WORK | 2025-03-29] Worked 8.5 hours
             └ [STRESS | 2025-03-29] Self-reported stress: 6/10
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RISK SUMMARY
  High risk days     : 2
  Moderate risk days : 1
  Low risk days      : 1
  Average score      : 6.5 / 12
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ANALYSIS & RECOMMENDATION
=== Weekly Analysis for Ahona Sarkar ===
Days analyzed: 4
Average stress score: 6.5
High-risk days: 2
Recommendation: Consider speaking to a counselor or doctor.
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```

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